APPLICATIONS OF COMPUTATIONAL FLUID DYNAMICS

UNIT – IV

Finite Volume Method

Syllabus (as given)

- Finite volume method via finite difference method.
- Formulations for two- and three-dimensional problems.

1. Basic Idea of the Finite Volume Method (FVM)

FVM = integrate the governing conservation equation over each small control volume (CV) of the grid, then convert volume integrals of divergence terms into surface integrals (fluxes) using Gauss' divergence theorem. The discrete equations express exact conservation of mass, momentum, energy over every CV.

$$\int_{CV} \nabla\cdot(\rho V\varphi)\, dV = \int_{A} (\rho V\varphi)\cdot n\, dA = \Sigma_{faces}\, F_f \varphi_f \quad \text{(Gauss theorem)}$$

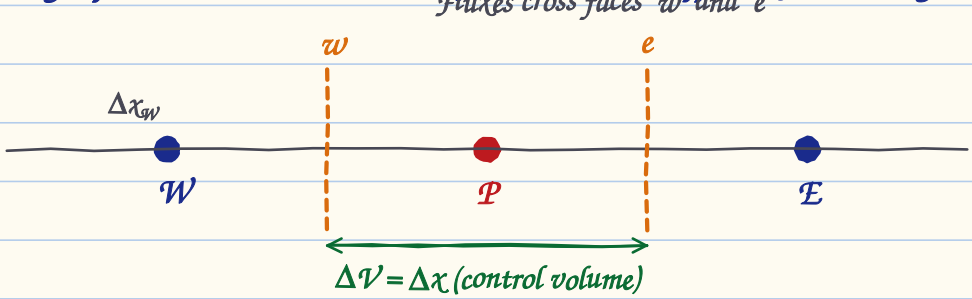
General transport equation

$$\partial(\rho\varphi)/\partial t + \nabla\cdot(\rho V\varphi) = \nabla\cdot(\Gamma\nabla\varphi) + S_\varphi$$

Integrate over CV and over time Δt:

$$\int_{\Delta t}[\partial/\partial t \int_{CV} \rho\varphi\, dV + \int_{A} n\cdot(\rho V\varphi)\, dA\,]\, dt = \int_{\Delta t}[\int_{A} n\cdot(\Gamma\nabla\varphi)\, dA + \int_{CV} S_\varphi\, dV\,]\, dt$$

i.e. (rate of change of φ in CV) + (net convective flux out) = (net diffusive flux in) + (generation).



Steps of FVM

1. Grid generation: divide domain into non-overlapping CVs; nodes P at CV centre; faces w, e (n, s, t, b).
2. Integrate the conservation equations over each CV.
3. Discretisation: approximate face values and gradients by interpolation (profile assumptions) -> algebraic equation per CV.
4. Assemble and solve algebraic equations (TDMA, Gauss–Seidel, multigrid); iterate for

non-linearities.

Why FVM?

- Conservative by construction (flux leaving one CV = flux entering neighbour).
- Works on structured, unstructured and non-orthogonal meshes; handles complex geometry.
- Physical interpretation of every term; basis of FLUENT, CFX, OpenFOAM, STAR-CD.

2. FVM for 1-D Steady Diffusion

Equation: $d/dx(\Gamma\, d\varphi/dx) + S = 0$ (e.g. heat conduction with $\Gamma = k$). Integrate over CV ($\Delta x = x_e - x_w$):

$$(\Gamma A\, d\varphi/dx)_e - (\Gamma A\, d\varphi/dx)_w + S\,\Delta V = 0$$

Face gradient by central (linear) profile between nodes: $(d\varphi/dx)_e = (\varphi_E - \varphi_P)/\delta x_{PE}$, $(d\varphi/dx)_w = (\varphi_P - \varphi_W)/\delta x_{WP}$. Source linearised: $S\Delta V = S_u + S_P\varphi_P$ ($S_P \le 0$).

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + b$$

Coefficient	Value	Meaning
a_W	$\Gamma_w A_w/\delta x_{WP}$	west conductance
a_E	$\Gamma_e A_e/\delta x_{PE}$	east conductance
a_P	$a_W + a_E - S_P$	centre coefficient
b	S_u	constant source

- Face conductivity for non-uniform Γ: harmonic mean $\Gamma_e = 2\Gamma_P\Gamma_E/(\Gamma_P + \Gamma_E)$ (keeps flux continuity across material interface); for uniform grid, linear interpolation $\Gamma_e = \tfrac{1}{2}(\Gamma_P + \Gamma_E)$.

Boundary nodes

- Dirichlet φ_B given at boundary: boundary-face gradient over half cell: $(\varphi_P - \varphi_B)/(\delta x/2)$ -> extra coefficient $a_B = 2\Gamma A/\delta x$: $S_u = a_B\varphi_B$, $S_P = -a_B$.
- Neumann (given flux q): add $q\,A$ to source term, set boundary a-coefficient to zero. Insulated: flux = 0.

Worked example — 1-D conduction in a rod

Rod $L = 0.5$ m, $A = 10^{-2}$ m², $k = 1000$ W/mK, $T_A = 100°C$, $T_B = 500°C$, no source. Use 5 CVs, $\Delta x = 0.1$ m.

Interior: $a_W = a_E = kA/\Delta x = 100$. Node 1: $a_W = 0$, $a_E = 100$, $S_P = -200$, $S_u = 200T_A$ -> $a_P = 300$. Node 5: $S_u = 200T_B$, $a_P = 300$.

Equations: $300T_1 = 100T_2 + 200(100)$; $200T_2 = 100T_1 + 100T_3$; … ; $300T_5 = 100T_4 + 200(500)$.

Solution (TDMA): $T = 140, 220, 300, 380, 460$ °C = exact line $T = 100 + 800x$ at $x = 0.05, 0.15, \ldots$ √

Source-term linearisation

Non-linear source $S(\varphi)$ is linearised about the current value φ^*: $S = S_C + S_P\varphi_P$ with $S_P \le 0$ so that a_P stays positive (diagonal dominance).

$$S(\varphi) \approx S^* + (dS/d\varphi)^*(\varphi_P - \varphi^*) \quad \rightarrow \quad S_P = (dS/d\varphi)^* \le 0\,, \quad S_C = S^* - (dS/d\varphi)^*\varphi^*$$

Case	Treatment
$S = 4 - 5\varphi$	$S_C = 4$, $S_P = -5$ (already linear, negative slope)
$S = 3 + 0.2\varphi$ (positive slope)	put all in $S_C = 3 + 0.2\varphi^*$ (explicit, lagged), $S_P = 0$
$S = x\varphi^2$ … (non-linear)	Newton: $S_P = -2x\varphi^*$, $S_C = x\varphi^{*2}$

3. FVM for 1-D Convection–Diffusion

Steady: $d(\rho u\varphi)/dx = d/dx(\Gamma\, d\varphi/dx)$, with continuity $d(\rho u)/dx = 0$. Define convective strength $F = \rho u$ and diffusion conductance $D = \Gamma/\delta x$ at faces:

$$F_e\varphi_e - F_w\varphi_w = D_e(\varphi_E - \varphi_P) - D_w(\varphi_P - \varphi_W)$$

Peclet number $Pe = F/D = \rho u \delta x/\Gamma$ (ratio of convection to diffusion). The remaining task: express φ_e, φ_w in terms of nodal values.

(a) Central differencing

$$\varphi_e = (\varphi_P + \varphi_E)/2\,, \quad \varphi_w = (\varphi_W + \varphi_P)/2$$

$$a_E = D_e - F_e/2\,, \quad a_W = D_w + F_w/2\,, \quad a_P = a_E + a_W + (F_e - F_w)$$

- Second-order accurate, but $a_E < 0$ when $Pe > 2$ -> unbounded, oscillating (wiggles) solution. Valid only for $|Pe| < 2$.

(b) Upwind differencing

$$\varphi_e = \varphi_P \ (\mathcal{F} > 0), \quad \varphi_e = \varphi_E \ (\mathcal{F} < 0)$$

$$a_W = \mathcal{D}_w + \max(\mathcal{F}_w, 0), \quad a_E = \mathcal{D}_e + \max(-\mathcal{F}_e, 0)$$

- Always positive coefficients -> bounded and stable for all Pe; only first-order accurate; false (numerical) diffusion $\Gamma_{false} \approx \rho u \Delta x/2$.

(c) Hybrid scheme (Spalding)

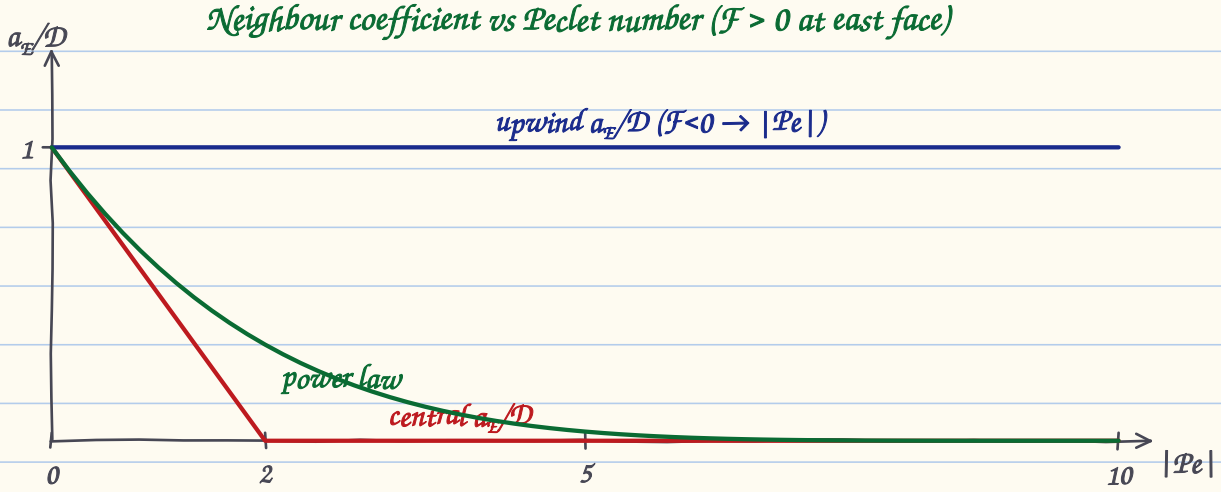
$$a_E = \max(-\mathcal{F}_e, \mathcal{D}_e - \mathcal{F}_e/2, 0), \quad a_W = \max(\mathcal{F}_w, \mathcal{D}_w + \mathcal{F}_w/2, 0)$$

- Central differencing for $|Pe| < 2$; upwind with diffusion set to zero for $|Pe| \geq 2$.

(d) Power-law and QUICK

- Power-law (Patankar): $a_E = \mathcal{D}_e \max(0, (1 - 0.1|Pe_e|)^5) + \max(-\mathcal{F}_e, 0)$ — close to the exact 1-D solution.
- QUICK (Leonard): quadratic upstream interpolation $\varphi_e = (6/8)\varphi_P + (3/8)\varphi_E - (1/8)\varphi_W$ $(\mathcal{F} > 0)$; third-order accurate, may be unbounded.

Neighbour coefficient vs Peclet number ($\mathcal{F} > 0$ at east face)



Three basic rules (Patankar)

- Conservativeness: flux through a common face is the same for both neighbouring CVs.
- Boundedness: all coefficients positive ($a_P = \Sigma a_{nb}$) — no overshoot.
- Transportiveness: influence of neighbours must depend on Pe (upstream influence grows with Pe).

Scheme	Order	Bounded	Remark
Central	2	$Pe < 2$ only	wiggles
Upwind	1	always	false diffusion
Hybrid	1	always	robust, simple
Power law	~1	always	accurate 1-D
QUICK	3	not always	needs limiter

Exact solution and worked example

Exact solution of the 1-D steady convection–diffusion problem with $\varphi = \varphi_0$ at $x = 0$ and $\varphi = \varphi_L$ at $x = L$:

$$(\varphi - \varphi_0)/(\varphi_L - \varphi_0) = (\exp(Pe \cdot x/L) - 1)/(\exp(Pe) - 1), \qquad Pe = \rho u L/\Gamma$$

Example: $\rho = 1, \Gamma = 0.1, L = 1$, 5 equal cells

$\delta x = 0.2$: $D = \Gamma/\delta x = 0.5$. $u = 0.1 \rightarrow F = 0.1$, $Pe_{cell} = F/D = 0.2 < 2$: central differencing is acceptable.

Interior coefficients (central): $a_W = D + F/2 = 0.55$, $a_E = D - F/2 = 0.45$, $a_P = a_W + a_E = 1.0$.

For $u = 2.5 \rightarrow F = 2.5$, $Pe_{cell} = 5 > 2$: central $a_E = 0.5 - 1.25 = -0.75 < 0$ (wiggles).

Upwind: $a_W = 0.5 + 2.5 = 3.0$, $a_E = 0.5$, $a_P = 3.5$ — bounded.

False diffusion

First-order upwind carries a truncation error that looks like extra diffusion $\Gamma_{num} \approx \rho u \Delta x(1 - \nu)/2$ ... for a flow oblique to the grid lines in 2-D $\Gamma_{num} \approx \rho |V| \Delta x \Delta y \sin\theta \cos\theta/(2(\Delta y |\sin\theta| + \Delta x |\cos\theta|))$. It is largest when the flow is at about 45° to the grid and gradients are steep; reduce by refining the grid or using higher-order bounded schemes (QUICK/limiters).

4. Unsteady Problems — Time Integration

1-D unsteady diffusion: $\rho c \partial T/\partial t = \partial/\partial x(k \partial T/\partial x) + S$. Integrate over CV and Δt:

$$\rho c \Delta x (T_P - T_P^0) = \int_t^{t+\Delta t} [(kA/\delta x)_e (T_E - T_P) - (kA/\delta x)_w (T_P - T_W)] \, dt$$

Weighting factor θ: $\int T \, dt = [\theta T + (1 - \theta) T^0] \Delta t$. $\theta = 0$ explicit, ½ Crank–Nicolson, 1 fully implicit.

$$a_P T_P = a_E T_E + a_W T_W + b, \quad a_P = a_E + a_W + a_P^0 \ (\theta = 1)$$

$$a_P^0 = \rho c \Delta x/\Delta t, \quad b = a_P^0 T_P^0 + S \Delta V$$

- **Explicit** ($\theta = 0$): $a_P = a_P^0$. Positive T_P^0 coefficient $a_P^0 - a_E - a_W \geq 0 \rightarrow \Delta t \leq \rho c (\Delta x)^2/2k$ (same as r

$\le \frac{1}{2}$).

- *Fully implicit* ($\theta = 1$): unconditionally stable and bounded, first order in time; used in most CFD codes. Crank–Nicolson: second order but may oscillate for large Δt.

Worked example: 2-D conduction on a 2×2 mesh

Square plate 1 m × 1 m, $k = 1$, no source

Top face $T = 100°C$, other three faces $0°C$. Four CVs ($\Delta x = \Delta y = 0.5$): A, B (top row), C, D (bottom row).

Interior conductance $a = k\Delta y/\Delta x = 1$; boundary (half cell) $a_B = 2k\Delta y/\Delta x = 2$. Cell A: $a_P = 1 + 1 + 2 + 2 = 6$.

$6A = B + C + 2(100) + 2(0)$; $6B = A + D + 200$; $6C = A + D$; $6D = B + C$.

By symmetry $A = B$, $C = D$: $5A - C = 200$, $5C = A$ -> $A = B = 41.7°C$, $C = D = 8.3°C$ (coarse mesh — refine to converge).

Line-by-line TDMA for 2-D

1. Choose a grid line (say row j). Treat φ_W, φ_P, φ_E on this line as unknowns.
2. Move N and S neighbours (latest values) to the right-hand side: $d_i = a_N\varphi_N + a_S\varphi_S + b$.
3. Solve the resulting tridiagonal system by TDMA ($a_i = -a_W$, $b_i = a_P$, $c_i = -a_E$).
4. Sweep the next row; repeat sweeps (alternating row and column directions) until convergence.

Unsteady convection–diffusion (explicit FVM)

$$\varphi_P^{n+1} = \varphi_P^n + (\Delta t/\rho\Delta x)[\,(D_e(\varphi_E - \varphi_P) - D_w(\varphi_P - \varphi_W)) - (F_e\varphi_e - F_w\varphi_w)\,]^n$$

- Stability: upwind explicit requires $\Delta t \le \rho\Delta x/(F + 2D)$ i.e. both the CFL ($\nu\Delta t/\Delta x \le 1$) and the diffusion restriction ($2\alpha\Delta t/\Delta x^2 \le 1$) combined: $\nu + 2r \le 1$.
- Implicit upwind has coefficients $a_P = a_E + a_W + a_P^0 - S_P\Delta x$ — positive and bounded for any Δt.

5. Finite Volume Method via Finite Difference Method

For a uniform Cartesian grid FVM and FDM are closely related. Start from a conservation law and treat the CV as a finite-difference cell with fluxes at half points $i \pm \frac{1}{2}$:

$$\partial u/\partial t + \partial f/\partial x = 0 \quad \text{->} \quad du_i/dt = -(f_{i+1/2} - f_{i-1/2})/\Delta x$$

Here u_i is the cell average $(1/\Delta x)\int u\,dx$ and $f_{i\pm1/2}$ are numerical fluxes at the faces — a finite-difference formula written in flux (conservative) form.

Equivalence with central differencing

- Diffusion: $(\Gamma\varphi_x)_e - (\Gamma\varphi_x)_w$ with linear profiles $= \Gamma(\varphi_E - 2\varphi_P + \varphi_W)/\Delta x$ -> identical to the

3-point FD formula $\Gamma\varphi_{xx}\Delta x$.

- Convection with central face values: $(F/2)(\varphi_E - \varphi_W)$ = FD central difference $\Delta x{\cdot}u_x F$.
- FDM works with point values and Taylor series; FVM works with cell averages and integral balance — FVM remains valid (and conservative) on non-uniform/unstructured grids, whereas FDM expressions need coordinate transformation.

Numerical flux functions — schemes in FVM form

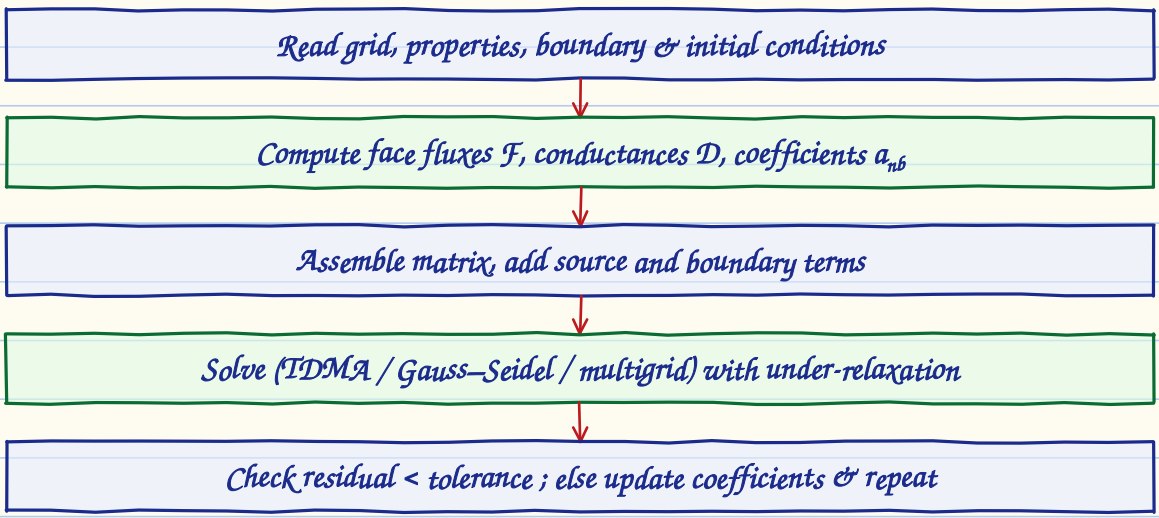
Scheme	Numerical flux $f_{i+1/2}$	Order
Central	$(f_i + f_{i+1})/2$	2
Upwind (a>0)	f_i	1
Lax–Friedrichs	$(f_i + f_{i+1})/2 - (\Delta x/2\Delta t)(u_{i+1} - u_i)$	1
Lax–Wendroff	$(f_i + f_{i+1})/2 - (\Delta t/2\Delta x)A_{i+1/2}(f_{i+1} - f_i)$	2
Godunov	exact Riemann solution at face	1
Roe	$\frac{1}{2}(f_L + f_R) - \frac{1}{2}\lvert A_{Roe}\rvert (u_R - u_L)$	1 (2 with MUSCL)

Update: $u_i^{n+1} = u_i^n - (\Delta t/\Delta x)(f_{i+1/2} - f_{i-1/2})$. Any FD scheme in conservative form can be viewed as an FVM; the choice of the face flux fixes accuracy, dissipation and shock-capturing ability.

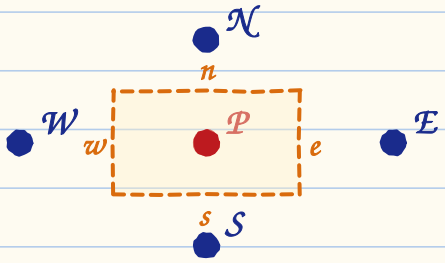
Higher-order reconstruction (MUSCL)

- Reconstruct linear profile inside each cell: $u^L_{i+1/2} = u_i + \frac{1}{2}\psi(r)(u_{i+1} - u_i)$, $u^R_{i+1/2} = u_{i+1} - \frac{1}{2}\psi(r)(u_{i+2} - u_{i+1})$ with limiter ψ -> second-order, TVD, non-oscillatory.

Algorithm of an FVM code



6. Two-dimensional Formulation



Cell P with neighbours W, E (east-west), N, S

faces w, e, n, s (dashed)

$\Delta V = \Delta x \cdot \Delta y \cdot 1$

Steady 2-D convection–diffusion: $\partial(\rho u\varphi)/\partial x + \partial(\rho v\varphi)/\partial y = \partial/\partial x(\Gamma \partial\varphi/\partial x) + \partial/\partial y(\Gamma \partial\varphi/\partial y) + S$.

Integrate over CV ($\Delta x \times \Delta y \times 1$):

$$(J_e - J_w)\Delta y + (J_n - J_s)\Delta x = S\Delta x\Delta y, \quad J = \rho u\varphi - \Gamma\, \partial\varphi/\partial x \ \text{(total flux)}$$

$$F_e = (\rho u)_e \Delta y, \ F_w = (\rho u)_w \Delta y, \ F_n = (\rho v)_n \Delta x, \ F_s = (\rho v)_s \Delta x$$

$$D_e = \Gamma_e \Delta y/\delta x_{PE}, \ D_w = \Gamma_w \Delta y/\delta x_{WP}, \ D_n = \Gamma_n \Delta x/\delta y_{PN}, \ D_s = \Gamma_s \Delta x/\delta y_{SP}$$

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + a_N\varphi_N + a_S\varphi_S + b$$

Coefficient	Upwind / hybrid-type value	Notes
a_E	$D_e + \max(-F_e, 0)$	east
a_W	$D_w + \max(F_w, 0)$	west
a_N	$D_n + \max(-F_n, 0)$	north
a_S	$D_s + \max(F_s, 0)$	south
a_P	$a_E + a_W + a_N + a_S + \Delta F - S_P\Delta x\Delta y$	$\Delta F = 0$ if continuity holds
b	$S_u\Delta x\Delta y$ (+ $a_P^0\varphi_P^0$ unsteady)	source

Pure 2-D diffusion (conduction, uniform grid)

$$a_E = a_W = k\Delta y/\Delta x, \quad a_N = a_S = k\Delta x/\Delta y, \quad a_P = a_E + a_W + a_N + a_S$$

- For $\Delta x = \Delta y = h$ and no source: $T_P = (T_E + T_W + T_N + T_S)/4$ — identical to the five-point finite-difference formula of Unit I.

Solving the 2-D equations

- Pentadiagonal sparse matrix. Gauss–Seidel / SOR point-by-point; or line-by-line TDMA: take unknowns along a line implicitly (tridiagonal) with neighbouring lines (N, S) lagged, sweep in alternating directions (ADI-like).
- Multigrid or conjugate-gradient methods for large grids. Under-relaxation for non-linear problems: $\varphi_P = \varphi_P^{old} + \alpha(\varphi_P^{new} - \varphi_P^{old})$.

- Boundary cells: boundary face fluxes added to source terms (Dirichlet: $a_B = 2\Gamma A/\delta$; Neumann: given flux).

7. Three-dimensional Formulation

Add top (t) and bottom (b) faces for a CV of size $\Delta x \times \Delta y \times \Delta z$. For steady convection–diffusion with continuity satisfied:

$$a_P\varphi_P = a_E\varphi_E + a_W\varphi_W + a_N\varphi_N + a_S\varphi_S + a_T\varphi_T + a_B\varphi_B + b$$

$$a_T = D_t + \max(-F_t, 0), \quad a_B = D_b + \max(F_b, 0), \quad D_t = \Gamma_t \Delta x\Delta y/\delta z_{PT}, \quad F_t = (\rho w)_t \Delta x\Delta y$$

$$a_P = a_E + a_W + a_N + a_S + a_T + a_B - S_P\Delta x\Delta y\Delta z$$

- Seven-diagonal (heptadiagonal) matrix; unsteady fully implicit adds $a_P^0 = \rho\Delta x\Delta y\Delta z/\Delta t$.
- Solvers: Gauss–Seidel with plane/line relaxation (TDMA along lines), ADI with Douglas–Gunn splitting, ICCG / BiCGSTAB, algebraic multigrid.
- Pure diffusion on uniform cube grid: $T_P = (T_E + T_W + T_N + T_S + T_T + T_B)/6$.

Non-orthogonal and unstructured meshes

- Face fluxes use area vectors: $\Sigma_f (\rho V\varphi)_f A_f$ and gradient $(\nabla\varphi)_f A_f = (\varphi_N - \varphi_P)|A_f|/d$·(correction terms for non-orthogonality).
- Gradient by Green–Gauss $(\nabla\varphi)_P = (1/V)\Sigma_f \varphi_f A_f$ or least squares; deferred correction keeps a compact stencil.
- Vertex-centred vs cell-centred storage; volumes may be tetrahedra, hexahedra, prisms, polyhedra.

Extension to the flow equations

- Momentum equations are convection–diffusion equations for u, v, w with pressure-gradient source — pressure-velocity coupling by SIMPLE family (Unit III) on a staggered or collocated grid (Rhie–Chow).
- For compressible flows, FVM of the Euler/NS equations: $\Sigma_{faces}(F\cdot n)_f A_f$ with Roe/AUSM/HLLC flux + MUSCL reconstruction + RK time marching.

Grid types

Grid	Description	Used with
Structured (Cartesian)	regular i,j,k indexing	FDM, simple FVM, ADI/TDMA
Body-fitted curvilinear	mapped to rectangle	FVM with metrics
Unstructured	triangles / tets / polyhedra	FVM, FEM
Hybrid / multi-block	blocks, prism layers	industrial FVM codes

- Grid stretching near walls ($\Delta y_{j+1} = r\Delta y_j$, $r \approx 1.05$-1.2) resolves boundary layers; first cell height from $y^+ \approx 1$ (wall-resolved).
- Cell skewness, aspect ratio, non-orthogonality control accuracy; use smooth variation of cell size.

Verification and grid independence

- Refine grid (e.g. h, $h/2$, $h/4$); observed order $p = \ln[(f_h - f_{h/2})/(f_{h/2} - f_{h/4})]/\ln 2$.
- Richardson extrapolation $f_{exact} \approx f_{h/2} + (f_{h/2} - f_h)/(2^p - 1)$; grid convergence index $GCI = 1.25|\varepsilon|/(r^p - 1)$.
- Validate against experiments/benchmarks (cavity, backward step, flat plate).

Short answer questions

Q1. What is a control volume? -> small cell around node P bounded by faces w, e, n, s over which conservation is enforced.

Q2. Why must $S_P \le 0$? -> to keep $a_P = \Sigma a_{nb} - S_P$ positive: diagonal dominance, stability and boundedness.

Q3. Peclet number? -> $Pe = F/D = \rho u \delta x/\Gamma$, ratio of convective to diffusive strength.

Q4. Why is upwind scheme stable but inaccurate? -> positive coefficients, but first-order false diffusion.

Q5. Difference between FDM and FVM? -> FDM uses Taylor series at points; FVM uses integral balances on cells; conservative on any mesh.

Q6. What is a collocated grid? -> all variables at cell centres; needs Rhie–Chow interpolation to avoid checkerboard pressure.

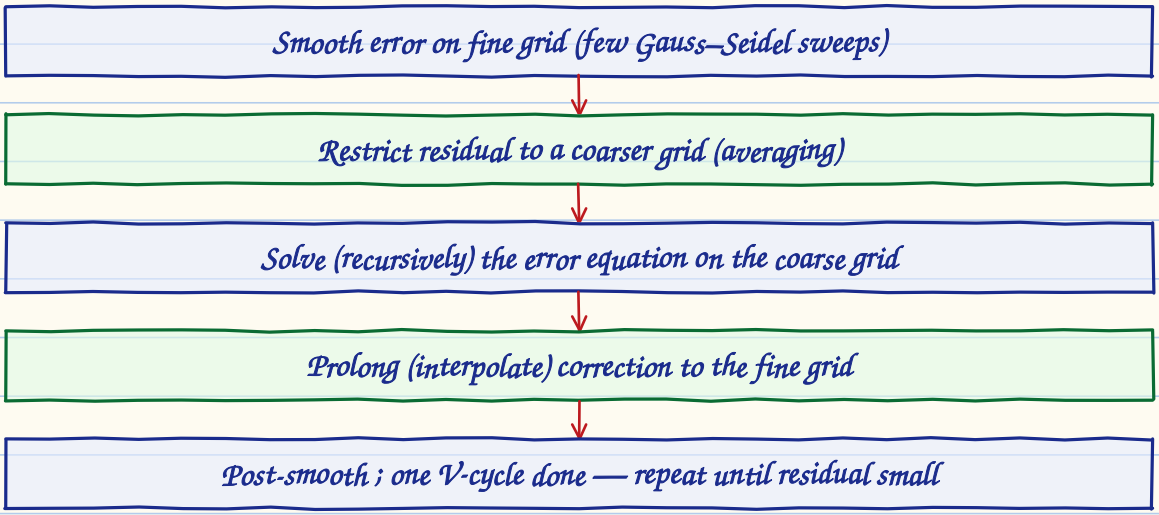
Pressure–velocity coupling in FVM: collocated grid and Rhie–Chow

If u, v, p are all stored at cell centres, a face velocity obtained by simple linear interpolation sees a pressure gradient that ignores neighbouring cells -> checkerboard p. Rhie–Chow adds a pressure-difference smoothing term at each face:

$$u_f = (\bar{u}_P + \bar{u}_E)/2 - d_f[\,(p_E - p_P)/\delta x - \tfrac{1}{2}((\nabla p)_P + (\nabla p)_E)\cdot e_x\,]$$

- The bracket is a fourth-order pressure smoothing term that couples neighbouring pressures; the mass flux built from u_f is used in the continuity (pressure-correction) equation of SIMPLE.

Multigrid acceleration (idea)



- Gauss–Seidel damps high-frequency error quickly but low-frequency error very slowly; coarse grids make low-frequency error look high-frequency. Work ≈ O(N) — much faster than SOR for large 3-D grids.

FVM for the 1-D Euler equations (Riemann-solver fluxes)

Each face has a left state U_L (cell i) and right state U_R (cell i+1): a local Riemann problem. The face flux is taken from its (approximate) solution:

$$U_i^{n+1} = U_i^n - (\Delta t/\Delta x)(F_{i+1/2} - F_{i-1/2}), \qquad F = F(U_L, U_R)$$

- Godunov: exact Riemann solution — accurate for shocks but expensive (iterative).
- Roe: linearised Riemann solver, $F = \tfrac{1}{2}[F(U_L) + F(U_R)] - \tfrac{1}{2}|\tilde{A}|(U_R - U_L)$; needs entropy fix for sonic points.
- HLL / HLLC: use only wave-speed estimates S_L, S_R — robust and simple; AUSM: splits the flux into convective and pressure parts.
- Time step restriction $\Delta t = CFL \cdot \Delta x / \max(|u| + a)$, $CFL \le 1$ for explicit RK/Euler.

Checklist: properties of a good FVM discretisation

Property	Meaning
Consistency	truncation error vanishes as $\Delta x, \Delta t \to 0$
Stability	errors do not amplify (explicit: CFL / diffusion limit)
Convergence	discrete solution → exact solution (Lax equivalence)
Conservativeness	global conservation = sum of local balances
Boundedness	solution stays between boundary / initial extremes
Transportiveness	upstream influence for convection-dominated flow
Accuracy	order, false diffusion, grid refinement behaviour

Axisymmetric and non-Cartesian coordinates

- CV volume and face areas change with r: $\Delta V = r_P \Delta r \Delta\theta \Delta z$, $A_e = r_E \Delta\theta \Delta z$ — coefficients $a_E = \Gamma\, r_P \Delta\theta \Delta z / \delta r$ etc.; metrics are absorbed in areas and volume, so the 1-D form $a_P \varphi_P = \Sigma a_{nb} \varphi_{nb} + b$ remains unchanged.

8. Summary and Important Questions

Quick revision — Unit IV

- FVM: integrate over CV -> flux balance -> $a_P \varphi_P = \Sigma a_{nb} \varphi_{nb} + b$.
- Diffusion: $a_{nb} = \Gamma A / \delta x$; convection–diffusion: central needs $Pe < 2$; upwind bounded but diffusive; hybrid/power-law/QUICK.
- FVM via FDM: $du_i/dt = -(f_{i+1/2} - f_{i-1/2})/\Delta x$; choose numerical flux.
- 2-D: five coefficients $a_{E,W,N,S,P}$; 3-D: add a_T, a_B (seven-point stencil).
- Solvers: TDMA line-by-line, Gauss–Seidel, multigrid.

1. Explain the finite volume method and derive the discretised equation for 1-D steady diffusion with boundary conditions.
2. Derive the discretisation of 1-D convection–diffusion by central, upwind and hybrid schemes; discuss Peclet number.
3. Show how the finite volume method is related to the finite difference method.
4. Formulate the FVM for 2-D and 3-D steady convection–diffusion problems.
5. Explain the implicit time-marching FVM for unsteady diffusion and its boundedness.